KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD

II PUC EXAM-2, APRIL-MAY 2026

Subject: 35-Mathematics SCHEME OF VALUATION MAX. MARKS: 80

Instructions:

a) Any answer by alternate method should be valued and suitably awarded.

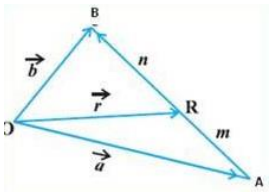
b) All answers (including extra, struck off and repeated) should be valued. Answers with maximum marks must be considered.

c) Units are not compulsory

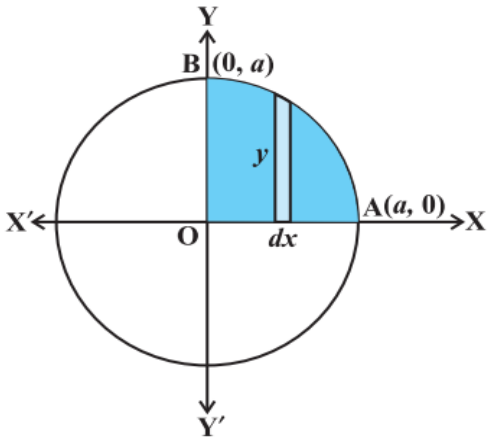
Qn No :	PART A	Marks
	I	
1	c) or writing Symmetric and Transitive	1
2	d) or writing $[0,\pi]-\left\{\frac{\pi}{2}\right\}$	1
3	b) or writing 2	1
4	a) or writing 4	1
5	c) or writing -1	1
6	a) or writing $2x$	1
7	d) or writing 126	1
8	d) or writing $a > 1$	1
9	a) or writing $x\,logx - x + C$	1
10	c) or writing 0	1
11	b) or writing Statement 1 is true and Statement 2 is false	1
12	c) or writing π	1
13	a) or writing $0, 0, -1$	1
14	b) or writing $\frac{1}{8}$	1
15	d) or writing $\frac{4}{9}$	1
	II	
16	-1	1
17	2	1
18	1	1
19	-2	1
20	0	1

PART B		
21	Taking $x = sin\theta$ and writing $\theta = sin^{-1}x$	1
	Proving the result $sin^{-1}(3x - 4x^3) = 3\ sin^{-1}x$	1
22	Writing Area = $\frac{1}{2}\begin{vmatrix}1 & 1 & 1\\1 & 6 & 4\\0 & 0 & 3\end{vmatrix}$ **OR** $\frac{1}{2}\begin{vmatrix}1 & 0 & 1\\6 & 0 & 1\\4 & 3 & 1\end{vmatrix}$	1
	Getting Area = $\frac{15}{2} sq\ units$	1
23	Getting $2 + 3\frac{dy}{dx} = cosy\ \frac{dy}{dx}$	1
	Writing $\frac{dy}{dx} = \frac{2}{cosy - 3}$	1
24	Writing Area, $A = \pi r^2$ and $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$	1
	Getting $\frac{dA}{dt} = 60\pi\ sq\ cm/sec$	1
25	Taking $tan^{-1}x = t$ and writing $\frac{1}{1+x^2}\ dx = dt$	1
	Getting $\int \frac{sin(tan^{-1}x)}{1+x^2}\ dx = -\cos(tan^{-1}x) + C$	1
26	Writing $(2 - y)\ dy = (x + 1)\ dx$	1
	Getting $2y - \frac{y^2}{2} = \frac{x^2}{2} + x + C$	1
27	Getting $\vec{a} \times \vec{b} = 5\hat{\imath} + \hat{\jmath} - 4\ \hat{k}$	1
	Getting area of parallelogram $\sqrt{42}$ sq. units	1
28	Writing $\overrightarrow{b_1} \cdot \overrightarrow{b_2} = 19$ **OR** $\lvert\overrightarrow{b_1}\rvert = 3$ and $\lvert\overrightarrow{b_2}\rvert = 7$	1
	Getting $\theta = cos^{-1}\frac{19}{21}$	1
29	Writing P(black ball in first draw) = P(E) $= \frac{10}{18} = \frac{5}{9}$ **OR** P(red ball in second draw) = P(F|E) $= \frac{8}{18} = \frac{4}{9}$	1
	Getting $P(E \cap F) = P(E) \times P(F\|E) = \frac{20}{81}$	1

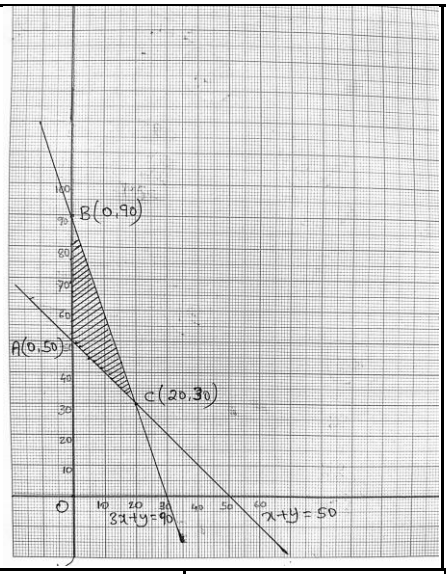
PART C		
30	Reflexive: $\forall L_1 \in L$, L_1 is parallel to L_1 **OR** Every line is parallel to itself $\Rightarrow (L_1, L_1) \in R$	1
	Symmetric: $let\ (L_1, L_2) \in R \Rightarrow$ L_1 is parallel to $L_2 \Rightarrow$ L_2 is parallel to L_1 $\Rightarrow (L_2, L_1) \in R$	1
	Transitive: $let\ (L_1, L_2) and\ (L_2, L_3) \in R \Rightarrow$ L_1 is parallel to L_2 & L_2 is parallel to L_3 $\Rightarrow$ L_1 is parallel to $L_3 \Rightarrow (L_1, L_3) \in R$ and writing conclusion.	1
31	Taking $cos^{-1}\left(\frac{4}{5}\right) = x$, $cos^{-1}\left(\frac{12}{13}\right) = y$, and getting $sinx = \frac{3}{5}$, $siny = \frac{5}{13}$	1
	Getting $cos(x + y) = \frac{33}{65}$	1
	Proving $cos^{-1}\frac{4}{5} + cos^{-1}\frac{12}{13} = cos^{-1}\frac{33}{65}$ **OR** for any other alternate method, allot appropriate marks	1
32	Writing P = $\frac{1}{2}$ [A + A^{I}]= $\begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix}$ is symmetric	1
	Writing Q = $\frac{1}{2}$ [A - A^{I}]= $\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ is skew symmetric	1
	Writing A = P + Q = $\begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$	1
33	Getting $\frac{dx}{d\theta} = a\ (1 - cos\theta)$	1
	Getting $\frac{dy}{d\theta} = a\ sin\theta$	1
	Getting $\frac{dy}{dx} = \frac{sin\theta}{(1-cos\theta)}$	1
34	Taking $x + y = 24$ and getting $f(x) = x(24 - x)$	1
	Getting $f'(x) = 24 - 2x$, $f''(x) = -2 < 0$	1
	Getting $x = 12$ & $y = 12$	1

35	Writing $\frac{x}{(x-1)(x-2)} = \frac{A}{(x-1)} + \frac{B}{(x-2)}$	1
	Getting: $A = -1$ and $B = 2$	1
	Getting $\int \frac{x}{(x-1)(x-2)} dx = 2\,log\lvert x-2\rvert - log\lvert x-1\rvert + C$ **OR** $\log \frac{(x-2)^2}{(x-1)} + C$	1
36	Writing the correct figure 	1
	Writing $\frac{m}{n} = \frac{AR}{RB}$ and $m \cdot \overrightarrow{RB} = n \cdot \overrightarrow{AR}$	1
	Getting $\vec{r} = \frac{m\,\vec{b} + n\,\vec{a}}{m+n}$	1
37	Writing $\overrightarrow{a_2} - \overrightarrow{a_1} = 2\hat{\imath} + \hat{\jmath} - \hat{k}$ **OR** $\lvert\vec{b}\rvert = 7$	1
	Getting $(\overrightarrow{a_2} - \overrightarrow{a_1}) \times \vec{b} = 9\hat{\imath} - 14\hat{\jmath} + 4\hat{k}$	1
	Getting distance, $d = \frac{\sqrt{293}}{7}$ units	1
38	Writing $P(E_1) = P(E_2) = \frac{1}{2}$	1
	Writing $P(A\lvert E_1) = \frac{3}{7}$, $P(A\lvert E_2) = \frac{5}{11}$	1
	Getting $P(E_2\lvert A) = \frac{35}{68}$	1
	PART D	
39	Writing $x_1, x_2 \in R$ *such that* $f(x_1) = f(x_2) \Rightarrow 10x_1 + 7 = 10x_2 + 7$	1
	Getting $x_1 = x_2$, and writing f is one-one.	1
	$y = 10x + 7 \Rightarrow x = \frac{y-7}{10} \in$ R	1
	$\forall\, y \in R$ *there exists* $x = \frac{y-7}{10} \in$ R such that $f(x) = y \Rightarrow$ f is onto	1
	Writing f is invertible and $f^{-1}(y) = \frac{y-7}{10}$ **OR** $f^{-1}(x) = \frac{x-7}{10}$	1

	OR	
	Getting $g(y)=\frac{y-7}{10}$	1
	Proving $g \circ f(x) = x$ °o$^{o0}{}_{o}$°°	1
	Proving $f \circ g\,(y) = y$	1
	Writing $g \circ f = f \circ g = I_R$ and f is invertible	1
	Writing $f^{-1}(y)=\frac{y-7}{10}$ **OR** $f^{-1}(x)=\frac{x-7}{10}$	1
40	Getting $AB=\begin{bmatrix}-2 & -6 & 12\\ 4 & 12 & -24\\ 5 & 15 & -30\end{bmatrix}$	1
	Writing $(AB)^T=\begin{bmatrix}-2 & 4 & 5\\ -6 & 12 & 15\\ 12 & -24 & -30\end{bmatrix}$	1
	Writing $A^T=[-2 \quad 4 \quad 5]$ and $B^T=\begin{bmatrix}1\\3\\-6\end{bmatrix}$	1
	Getting $B^TA^T=\begin{bmatrix}-2 & 4 & 5\\ -6 & 12 & 15\\ 12 & -24 & -30\end{bmatrix}$	1
	Concluding $(AB)^T=B^TA^T$	1
41	Writing AX=B $\Rightarrow \begin{bmatrix}1 & -1 & 2\\ 3 & 4 & -5\\ 2 & -1 & 3\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}7\\-5\\12\end{bmatrix}$ **OR** Writing A $=\begin{bmatrix}1 & -1 & 2\\ 3 & 4 & -5\\ 2 & -1 & 3\end{bmatrix}$, B $=\begin{bmatrix}7\\-5\\12\end{bmatrix}$, X $=\begin{bmatrix}x\\y\\z\end{bmatrix}$ **OR** Getting $\lvert A\rvert = 4 \neq 0$ **Note:** Award a mark, if student writes directly $\lvert A\rvert$ = 4	1
	Getting $adjA=\begin{bmatrix}7 & 1 & -3\\ -19 & -1 & 11\\ -11 & -1 & 7\end{bmatrix}$ **Note:** If any 4 cofactors are correct award 1 mark	2
	Writing $X = A^{-1}B=\frac{1}{4}\begin{bmatrix}7 & 1 & -3\\ -19 & -1 & 11\\ -11 & -1 & 7\end{bmatrix}\begin{bmatrix}7\\-5\\12\end{bmatrix}$	1
	Getting $x = 2,\ y = 1,\ z = 3$	1

42	Getting $y_1 = -3\sin(\log x)\frac{1}{x} + 4\cos(\log x)\frac{1}{x}$	1
	Writing $x\, y_1 = -3\sin(\log x) + 4\cos(\log x)$	1
	Getting $x\, y_2 + y_1 = -3\cos(\log x)\frac{1}{x} - 4\sin(\log x)\frac{1}{x}$	1
	Writing $x\, y_2 + y_1 = -\frac{1}{x}\, y$	1
	Getting $x^2 y_2 + x y_1 + y = 0$	1
43	Taking $x = a\ sin\theta$ and getting $dx = a\ cos\theta\ d\theta$	1
	Getting $\int \frac{1}{\sqrt{a^2 - x^2}}\, dx = \int 1\ d\theta$	1
	Getting $\int \frac{1}{\sqrt{a^2 - x^2}}\, dx = sin^{-1}\left(\frac{x}{a}\right) + c$	1
	Writing $\int \frac{1}{\sqrt{9 - 25x^2}}\, dx = \int \frac{1}{\sqrt{3^2 - (5x)^2}}\, dx$ **OR** $\frac{1}{5}\int \frac{1}{\sqrt{\left(\frac{3}{5}\right)^2 - x^2}}\, dx$	1
	Getting $\int \frac{1}{\sqrt{9 - 25x^2}}\, dx = \frac{1}{5}\ sin^{-1}\left(\frac{5x}{3}\right) + c$	1
44	Writing correct figure 	1
	Writing Area, $A = 4\int_0^a y\ dx$	1
	Writing $A = 4\int_0^a \sqrt{a^2 - x^2}\ dx$	1
	Getting $A = 4\left[\frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\ sin^{-1}\left(\frac{x}{a}\right)\right]_0^a$	1
	Getting Area = πa^2 square units	1

45	Writing $\frac{dy}{dx} - \frac{1}{x}y = 2x$	1
	Getting $P = -\frac{1}{x}$ and $Q = 2x$	1
	Getting IF $= e^{\int P\,dx} = \frac{1}{x}$	1
	Writing $y\frac{1}{x} = \int 2x\left(\frac{1}{x}\right)dx + C$	1
	Getting $\frac{y}{x} = 2x + C$ **OR** $y = 2x^2 + Cx$ (Deduct 1 mark for not writing C)	1
	PART E	
46	Taking $x = a - t$ **OR** $a - x = t \Rightarrow dx = -dt$ $x = 0 \Rightarrow$ t = a and x = a $\Rightarrow$ t = 0	1
	Getting $\int_0^a f(x)\,dx = \int_a^0 f(a-t)\,(-dt)$	1
	Getting $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$	1
	Getting $I = \int_0^4 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{4-x}}dx = \int_0^4 \frac{\sqrt{4-x}}{\sqrt{4-x}+\sqrt{x}}dx$	1
	Getting $2I = \int_0^4 1\,dx$	1
	Getting $I = 2$	1
	OR	

		Drawing line x + y = 50 correctly Drawing line 3x + y = 90 correctly For shading the feasible region	1 1 1
		Corner points / $Z = 4x + y$ (0, 50) — 50 (0, 90) — 90 (20, 30) — 110	1 + 1
	Writing the conclusion minimum value of Z is 50 at $(0, 50)$ and maximum value of Z is 110 at $(20, 30)$		1
47	Getting $A^2 = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$		1
	Proving $A^2 - 5A + 7I = O$		1
	Getting $7A^{-I} = 5I - A$ ***OR*** $A^{-I} = \frac{1}{7}(5I - A)$		1
	Getting $A^{-1} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$		1
	OR		
	Writing $\lim_{x\to\pi^-} f(x) = \lim_{x\to\pi^+} f(x) = f(\pi)$ ***OR*** LHL = RHL		1
	Writing $\lim_{x\to\pi^-} f(x) = k\pi + 1$ ***OR*** $f(\pi) = k\pi + 1$		1
	Writing $\lim_{x\to\pi^+} f(x) = cos\,\pi$ ***OR*** -1		1
	Getting $k = -\frac{2}{\pi}$		1